UNIT-3 (DISCRETE MATHEMATICAL STRUCTURE)

COMBINATORICS

LECTURE NO:-19

Combinatorics

combinatorics, also called combinatorial mathematics, the field of mathematics concerned with problems of selection, arrangement, and operation within a finite or discrete system. Included is the closely related area of combinatorial geometry.

One of the basic problems of combinatorics is to determine the number of possible configurations (e.g., graphs, designs, arrays) of a given type. Even when the rules specifying the configuration are relatively simple, enumeration may sometimes present formidable difficulties. The mathematician may have to be content with finding an approximate answer or at least a good lower and upper bound.

In mathematics, generally, an entity is said to "exist" if a mathematical example satisfies the abstract properties that define the entity. In this sense it may not be apparent that even a single configuration with certain specified properties exists. This situation gives rise to problems of existence and construction. There is again an important class of theorems that guarantee the existence of certain choices under appropriate hypotheses. Besides their intrinsic interest, these theorems may be used as existence theorems in various combinatorial problems.

Basic Counting Principles

Sum Rule Principle: Assume some event E can occur in m ways and a second event F can occur in n ways, and suppose both events cannot occur simultaneously. Then E or F can occur in m + n ways.

In general, if there are n events and no two events occurs in same time then the event can occur in n_1+n_2.....n ways.

Example: If 8 male processor and 5 female processor teaching DMS then the student can choose professor in 8+5=13 ways.

Product Rule Principle: Suppose there is an event E which can occur in m ways and, independent of this event, there is a second event F which can occur in n ways. Then combinations of E and F can occur in mn ways.

In general, if there are n events occurring independently then all events can occur in the order indicated as n_1 x n_2 x n_3.....n ways.

Example: In class, there are 4 boys and 10 girls if a boy and a girl have to be chosen for the class monitor, the students can choose class monitor in 4 x 10 = 40 ways.

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Mathematical Functions:

Factorial Function: The product of the first n natural number is called factorial n. It is denoted by n!, read "n Factorial."

The Factorial n can also be written as

1. n! = n (n-1) (n-2) (n-3).....1.
2. = 1 and 0! = 1.

Example1: Find the value of 5!

Solution:

5! = 5 x (5-1) (5-2) (5-3) (5-4)

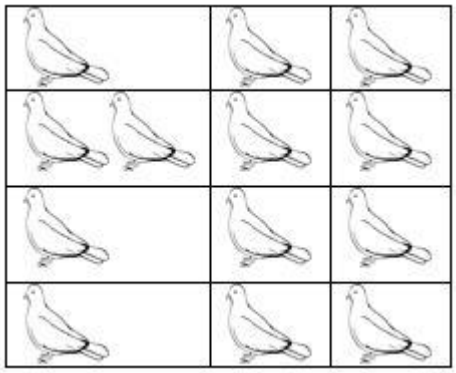
= 5 x 4 x 3 x 2 x 1 = 120

Example2: Find the value of $\frac{10!}{8!.}$

Solution: $\frac{10!}{8!.} = \frac{10 \times 9 \times 8!}{8!}$ = 10 x 9=90

The Pigeonhole Principal

Suppose that a flock of 20 pigeons flies into a set of 19 pigeonholes to roost. Because there are 20 pigeons but only 19 pigeonholes, at least one of these 19 pigeonholes must have at least two pigeons in it. To see why this is true, note that if each pigeonhole had at most one pigeon in it, at most 19 pigeons, one per hole, could be accommodated. This illustrates a general principle called the pigeonhole principle, which states that if there are more pigeons than pigeonholes, then there must be at least one pigeonhole with at least two pigeons in it.



Theorem – I) If "A" is the average number of pigeons per hole, where A is not an integer then

At least one pigeon hole contains **ceil[A]** (smallest integer greater than or equal to A) pigeons
Remaining pigeon holes contains at most **floor[A]** (largest integer less than or equal to A) pigeons

Or **II)** We can say as, if n + 1 objects are put into n boxes, then at least one box contains two or more objects. The abstract formulation of the principle: Let X and Y be finite sets and let $f : A \to B$ be a function.

- If X has more elements than Y, then f is not one-to-one.
- If X and Y have the same number of elements and f is onto, then f is one-to-one.
- If X and Y have the same number of elements and f is one-to-one, then f is onto.

Pigeonhole principle is one of the simplest but most useful ideas in mathematics. We will see more applications that proof of this theorem.

Example – 1: If (Kn+1) pigeons are kept in n pigeon holes where K is a positive integer, what is the average no. of pigeons per pigeon hole?
Solution: average number of pigeons per hole = (Kn+1)/n = K + 1/n Therefore there will be at least one pigeonhole which will contain at least (K+1) pigeons i.e., ceil[K +1/n] and remaining will contain at most K i.e., floor[k+1/n] pigeons. i.e., the minimum number of pigeons required to ensure that at least one pigeon hole contains (K+1) pigeons is (Kn+1).
Example – 2: A bag contains 10 red marbles, 10 white marbles, and 10 blue marbles. What is the minimum no. of marbles you have to choose randomly from the bag to ensure that we get 4 marbles of same color?
Solution: Apply pigeonhole principle. No. of colors (pigeonholes) n = 3 No. of marbles (pigeons) K+1 = 4 Therefore the minimum no. of marbles required = Kn+1 By simplifying we get Kn+1 = 10. Verification: ceil[Average] is [Kn+1/n] = 4 [Kn+1/3] = 4 Kn+1 = 10 i.e., 3 red + 3 white + 3 blue + 1(red or white or blue) = 10

Pigeonhole principle strong form – Theorem: Let q1, q2, . . . , qn be positive integers. If q1+ q2+ . . . + qn – n + 1 objects are put into n boxes, then either the 1st box contains at least q1 objects, or the 2nd box contains at least q2 objects, . . ., the nth box contains at least qn objects.

Example – 1: In a computer science department, a student club can be formed with either 10 members from first year or 8 members from second year or 6 from third year or 4 from final year. What is the minimum no. of students we have to choose randomly from department to ensure that a student club is formed?
Solution: we can directly apply from the above formula where, q1 =10, q2 =8, q3 =6, q4 =4 and n=4 Therefore the minimum number of students required to ensure department club to be formed is 10 + 8 + 6 + 4 – 4 + 1 = 25
Example – 2: A box contains 6 red, 8 green, 10 blue, 12 yellow and 15 white balls. What is the minimum no. of balls we have to choose randomly from the box to ensure that we get 9 balls of same color?
Solution: Here in this we cannot blindly apply pigeon principle. First we will see what happens if we apply above formula directly. From the above formula we have get answer 47 because 6 + 8 + 10 + 12 + 15- 5 + 1 = 47 But it is not correct. In order to get the correct answer we need to include only blue, yellow and white balls because red and green balls are less than 9. But we are picking randomly so we include after we apply pigeon principle. i.e., 9 blue + 9 yellow + 9 white – 3 + 1 = 25 Since we are picking randomly so we can get all the red and green balls

before the above 25 balls. Therefore we add 6 red + 8 green + 25 = 39 We can conclude that in order to pick 9 balls of same color randomly, one has to pick 39 balls from a box.

References:

1. Kenneth H. Rosen. Discrete Mathematics and its applications. Tata McGraw Hill.

2. Kolman, Busby and Ross, Distrete Mathematics Structures, PHI.

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UNIT-3 (DISCRETE MATHEMATICAL STRUCTURE)

PERMUTATIONS AND COMBINATIONS AND BINOMIAL COEFFICIENTS

LECTURE NO:-20

Permutations and combinations and Binomial Coefficients

Permutation and Combination: Permutation and Combination are one of the most important concepts in Mathematics. It is used in the practical applications in science, engineering and research, and many other fields. The difference between permutation and combination can be defined as; when a set of data is selected from a certain group, it is known as permutation; while the order in which the data is arranged is known as combination. Thus, we can say that permutation is an ordered combination.

Grouping of data and its interpretation is possible with the help of permutation and combination. Mathematics students should be well versed in the concepts because it finds applications in all advanced research and data analysis. This article provides insight on the permutation and combination questions, solutions, permutation and combination formulas and more.

A permutation is an arrangement in a definite order of several objects taken, some or all at a time, with permutations, every tiny detail matters. It means the order in which elements are arranged is significant.

There are two types of permutations:

- **Repetition is Allowed:** For the number lock example provided above, it could be “2-2-2”.
- **No Repetition Allowed:** For example, the first three people in a race. You can’t be first and second at the same time.

The combination is a way of selecting elements from a set so that the order of selection doesn’t matter. With the combination, only choosing elements matters. It means the order in which elements are chosen is not essential.

There are two types of combinations:

- **Repetition is Allowed**: For example, coins in your pocket (2, 5, 5, 10, 10)
- **No Repetition Allowed**: For example, lottery numbers (2, 14, 18, 25, 30, 38)

Permutation and Combination Formula

There are many formulas that are used to solve permutation and combination problems. We have provided the complete permutation formula list here:

- **When repetition is not allowed:** P is a permutation or arrangement of r things from a set of n things without replacement.

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- **When repetition is allowed:** P is a permutation or arrangement of r things from a set of n things when duplication is allowed.

  Derivation of Permutation Formula: Let us assume that there are r boxes, and each of them can hold one thing. There will be as many permutations as there are ways of filling in r vacant boxes by n objects.

- No. of ways the first box can be filled: n
- No. of ways the second box can be filled: **(**n **– 1)**
- No. of ways the third box can be filled: **(n – 2)**
- No. of ways the fourth box can be filled: **(n – 3)**
- No. of ways rth box can be filled: **[**n **–** (r **– 1)]**
- The number of permutations of n different objects taken r at a time, where 0 < r ≤ n and the objects do not repeat is: **n(n – 1)(n – 2)(n – 3) . . . (n – r + 1)**

  We have provided the complete combination formula list here:

- **When repetition is not allowed:** C is a combination of n distinct things taking r at a time (order is not important).
- **When repetition is allowed:** C is a combination of n distinct things taking r at a time (order is not important) with repetition. We define C as:

Q.1: Find the number of permutations and combinations, if n = 15 and r = 3.
Ans: n = 15, r = 3 (Given)
Using the formulas for permutation and combination, we get:
Permutation, P = n!/(n – r)!
= 15!/(15 – 3)!
= 15!/12!
= (15 x 14 x 13 x 12!)/12!
= 15 x 14 x 13
= 2730
Also, Combination, C = n!/(n – r)!r!
= 15!/(15 – 3)!3!
= 15!/12!3!
= (15 x 14 x 13 x 12!)/12!3!
= 15 x 14 x 13/6
= 2730/6
= 455

Q.2: In how many different ways can the letters of the word 'OPTICAL' be arranged so that the vowels always come together?
Ans: The word 'OPTICAL' has 7 letters. It has the vowels' O', 'I', and 'A' in it and these 3 vowels should always come together. Hence these three vowels can be grouped and considered as a single letter. That is, PTCL(OIA).
Hence we can assume the total letters as 5 and all these letters are different.
Number of ways to arrange these letters
= 5!
= 5×4×3×2×1
= 120
All the 3 vowels (OIA) are different.

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Number of ways to arrange these vowels among themselves
= 3!
= 3×2×1
= 6
Hence, the required number of ways:
= 120×6
= 720

Q.3: How many 3-letter words with or without meaning, can be formed out of the letters of the word, 'LOGARITHMS' if repetition of letters is not allowed?
Ans: The word 'LOGARITHMS' has 10 different letters.
Hence, the number of 3-letter words (with or without meaning) formed by using these letters
= 10P3
= 10×9×8
= 720

Q.4: There are 8 men and 10 women and you need to form a committee of 5 men and 6 women. In how many ways can the committee be formed?
Ans: We need to select 5 men from 8 men and 6 women from 10 women.
Number of ways to do this
= 8C5 × 10C6
= 8C3 × 10C4 [∵ nCr = nC(n-r)]

= [(8 x 7 x 6)/(3 x 2 x 1)] x [(10 x 9 x 8 x 7)/(4 x 3 x 2 x 1)]

= 56×210
= 11760

Q.5: A bag contains 2 white balls, 3 black balls and 4 red balls. In how many ways can 3 balls be drawn from the bag if at least one black ball is to be included in the draw?
Ans: From 2 white balls, 3 black balls and 4 red balls, 3 balls are to be selected such that at least one black ball should be there.
Hence we have 3 choices as given below.
Choice 1: We can select 3 black balls.
Choice 2: We can select 2 black balls and 1 non-black ball.
Choice 3: We can select 1 black ball and 2 non-black balls.

Number of ways to select 3 black balls
= 3C3
Number of ways to select 2 black balls and 1 non-black ball
= 3C2 × 6C1
Number of ways to select 1 black ball and 2 non-black balls
= 3C1 × 6C2
Total number of ways
= 3C3 + 3C2 × 6C1 + 3C1 × 6C2
= 3C3 + 3C1 × 6C1 + 3C1 × 6C2[∵ nCr = nC(n-r)]

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= 1 + (3×6) + [3 x (6×5)/(2×1)]

= 1 + 18 + 45
= 64

Q.6: An event manager has ten patterns of chairs and eight patterns of tables. In how many ways can he make a pair of tables and chairs?
Ans: The event manager has 10 patterns of chairs and 8 pattern tables.
A chair can be selected in 10 ways.
A table can be selected in 8 ways.
Hence one chair and one table can be selected in 10 × 8 ways
= 80 ways

Q.7: In how many ways can three boys can be seated on five chairs?
Ans: There are 3 boys.
The 1st boy can sit in any of the five chairs (5 ways).

5

Now there are 4 chairs remaining. The second boy can sit in any of the four chairs (4 ways).

5	4

Now there are 3 chairs remaining. The third boy can sit in any of the three chairs (3 ways).

5	4	3

Hence, the total number of ways in which 3 boys can be seated on 5 chairs
= 5 × 4 × 3
= 60

Q.8: In how many ways can a team of 5 persons be formed out of a total of 10 persons such that two particular persons should be included in each team?
Answer: Two particular persons should be included in each team. Therefore we have to select the remaining (5 – 2) = 3 persons from (10 – 2) = 8 persons.
Hence, the required number of ways
= 8C3
= (8×7×6)/(3×2×1)
= 8×7
= 56

Introduction to the Binomial Theorem

The Binomial Theorem is the method of expanding an expression that has been raised to any finite power. A binomial Theorem is a powerful tool of expansion, which has application in Algebra, probability, etc.

Binomial Expression: A binomial expression is an algebraic expression that contains two dissimilar terms. Ex: a + b, a3 + b3, etc.

Binomial Theorem: Let $n \in N, x, y, \in R$ then

$(x + y)n = n\Sigma r=0\ nCr\ xn – r \cdot yr$ where,

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$$nC_r = \frac{n!}{(n-r)!r!}$$

Illustration 1: Expand (x/3 + 2/y)4

Sol:

$$\left(\frac{x}{3}+\frac{2}{y}\right)^4 = 4c_0\left(\frac{x}{3}\right)^4 + 4c_1\left(\frac{x}{3}\right)^3\left(\frac{2}{y}\right) + 4c_2\left(\frac{x}{3}\right)^2\left(\frac{2}{y}\right)^2 + 4c_3\left(\frac{x}{3}\right)\left(\frac{2}{y}\right)^3 + 4c_4\left(\frac{2}{y}\right)^4$$

$$\Rightarrow \frac{x^4}{81} + \frac{8x^3}{27y} + \frac{8x^2}{3y^2} + \frac{32x}{3y^3} + \frac{16}{y^4}$$

To find binomial coefficients we can also use **Pascal's Triangle**.

Pascal's Triangle

1

1 1

1 2 1

1 3 3 1

1 4 6 4 1

1 5 10 10 5 1

1 6 15 20 15 6 1

Finding binomial coefficients with Pascal's Triangle

Some other useful expansions:

- (x + y)n + (x−y)n = 2[C0 xn + C2 xn-1 y2 + C4 xn-4 y4 + …]
- (x + y)n − (x−y)n = 2[C1 xn-1 y + C3 xn-3 y3 + C5 xn-5 y5 + …]
- (1 + x)n = nΣr-0 nCr . xr = [C0 + C1 x + C2 x2 + … Cn xn]
- (1+x)n + (1 − x)n = 2[C0 + C2 x2+C4 x4 + …]
- (1+x)n − (1−x)n = 2[C1 x + C3 x3 + C5 x5 + …]
- The number of terms in the expansion of (x + a)n + (x−a)n are (n+2)/2 if "n" is even or (n+1)/2 if "n" is odd.
- The number of terms in the expansion of (x + a)n − (x−a)n are (n/2) if "n" is even or (n+1)/2 if "n" is odd.

Properties of Binomial Coefficients

Binomial coefficients refer to the integers which are coefficients in the binomial theorem. Some of the most important properties of binomial coefficients are:

- C0 + C1 + C2 + … + Cn = 2n

- C0 + C2 + C4 + … = C1 + C3 + C5 + … = 2n-1
- C0 – C1 + C2 – C3 + … +(−1)n . nCn = 0
- nC1 + 2.nC2 + 3.nC3 + … + n.nCn = n.2n-1
- C1 − 2C2 + 3C3 − 4C4 + … +(−1)n-1 Cn = 0 for n > 1
- C02 + C12 + C22 + …Cn2 = [(2n)!/ (n!)2]

Illustration: If (1 + x)15 = a0 + a1x + ... + a15 x15 then, find the value of $\sum_{r=1}^{15} r.\frac{ar}{a_{r-1}}$

Sol:

$$\sum_{r=1}^{15} r.\frac{ar}{a_{r-1}} = 1.\frac{a_1}{a_0} + 2.\frac{a_2}{a_1} + 3.\frac{a_3}{a_2} + \ldots + 15.\frac{a_{15}}{a_{14}}$$

= C1/C0 + 2 C2/C1+ 3C3/C2 + ... + 15 C15/C14

= 15 + 14 + 13 + ... + 1 = [15(15+1)]/2 = 120

General Term in binomial expansion:

We have (x + y)n = nC0 xn + nC1 xn-1 . y + nC2 xn-2 . y2 + … + nCn yn

General Term = Tr+1 = nCr xn-r . yr

- General Term in (1 + x)n is nCr xr
- In the binomial expansion of (x + y)n , the rth term from end is (n – r + 2)th .

Illustration: Find the number of terms in (1 + 2x +x2)50

Sol:

(1 + 2x + x2)50 = [(1 + x)2]50 = (1 + x)100

The number of terms = (100 + 1) = 101

Illustration: Find the fourth term from the end in the expansion of (2x – 1/x2)10

Sol:

Required term =T10 – 4 + 2 = T8 = 10C7 (2x)3 (−1/x2)7 = −960x-11

Middle Term(S) in the expansion of (x+y) n.n

- If n is even then (n/2 + 1) Term is the middle Term.
- If n is odd then [(n+1)/2]th and [(n+3)/2)th terms are the middle terms.

Illustration: Find the middle term of (1 −3x + 3x2 – x3)2n

Sol:

(1 − 3x + 3x2 – x3)2n = [(1 − x)3]2n = (1 − x)6n

Middle Term = [(6n/2) + 1] term = 6nC3n (−x)3n

Determining a Particular Term:

- In the expansion of(axp + b/xq)n the coefficient of xm is the coefficient of Tr+1 where r = [(np−m)/(p+q)]
- In the expansion of (x + a)n, Tr+1/Tr = (n – r + 1)/r . a/x

Independent Term

The term Independent of in the expansion of [axp + (b/xq)]n is

Tr+1 = nCr an-r br, where r = (np/p+q) (integer)

Illustration: Find the independent term of x in (x+1/x)6

Sol:

r = [6(1)/1+1] = 3

The independent term is 6C3 = 20

References:

1. Kenneth H. Rosen. Discrete Mathematics and its applications. Tata McGraw Hill.

2. Kolman, Busby and Ross, Distrete Mathematics Structures, PHI.

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UNIT-3 (DISCRETE MATHEMATICAL STRUCTURE)

INTRODUCTION TO RECURRENCE RELATION AND RECURSIVE ALGORITHMS

LECTURE NO:-21

Introduction to Recurrence Relation and Recursive Algorithms:

A **recurrence relation** is an equation which represents a sequence based on some rule. It helps in finding the subsequent term (next term) dependent upon the preceding term (previous term). If we know the previous term in a given series, then we can easily determine the next term. Since a standard pattern is developed now, we can find the set of new terms. This is also applicable for arithmetic and geometric sequence.

Recurrence Relation Definition

When we speak about a standard pattern, all the terms in the relation or equation have the same characteristics. It means if there is a value of 'n', it can be used to determine the other values by just entering the value of 'n'.

The value of n should be organised and accurate, which is known as the Simplest form. In case of the simplest form of any such relation, the next term is dependent only upon the previous term. The sequence or series generated by recurrence relation is called a **Recurrence Sequence**.

Recurrence Relation Formula

Let us assume xn is the nth term of the series. Then the recurrence relation is shown in the form of;

$$x_n + 1 = f(x_n) \; ; \; n>0$$

Where f(xn) is the function.

We can also define a recurrence relation as an expression that represents each element of a series as a function of the preceding ones.

$$x_n = f(n, x_{n-1}) \; ; \; n>0$$

To write the recurrence relation of first-order, say order k, the above formula can be represented as;

$$x_n = f(n, x_{n-1}, x_{n-2}, \ldots\ldots, x_{n-k}) \; ; \; n-k>0$$

Examples of Recurrence Relation

In Mathematics, we can see many examples of recurrence based on series and sequence pattern. Let us see some of the examples here.

Factorial Representation

We can define the factorial by using the concept of recurrence relation, such as;

n!=n(n-1)! ; n>0

When n = 0,

0! = 1 is the initial condition.

To find the further values we have to expand the factorial notation, where the succeeding term is dependent on the preceding one.

Fibonacci Numbers

In Fibonacci numbers or series, the succeeding terms are dependent on the last two preceding terms. Therefore, this series is the best example of recurrence. As we know from the definition of the Fibonacci sequence,

Fn = Fn-1 + Fn-2

Now, if we take the initial values;

F0 = 0 and F1 = 1

So, F2 = F1 + F0 = 0 + 1 = 1

In the same way, we can find the next succeeding terms, such as;

F3 = F2 + F1

F4 = F3 + F2

And so on.

Thus, the Fibonacci series is given by;

0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, …∞

In the same way, there are other examples of recurrence such as a logical map, binomial coefficients where the same concept is applicable. Also, arithmetic and geometric series could be called a recurrence sequence.

Solving Recurrence Relations

To solve given recurrence relations we need to find the initial term first. Suppose we have been given a sequence;

an = 2an-1 – 3an-2

- Now the first step will be to check if initial conditions a0 = 1, a1 = 2, gives a closed pattern for this sequence.
- Then try with other initial conditions and find the closed formula for it.
- The result so obtained after trying different initial condition produces a series.
- Check the difference between each term, it will also form a sequence.

- We need to add all the terms of the new sequence, to understand which sequence is formed
- After understanding the pattern we can now identify the initial condition of the recurrence relation.

Recurrence Relation Problem

Now let us solve a problem based on the solution provided above.

Question: Solve the recurrence relation an = an-1 – n with the initial term a0 = 4.

Solution: Let us write the sequence based on the equation given starting with the initial number.

The sequence will be 4,5,7,10,14,19,.....

Now see the difference between each term.

a1 – a0 = 1

a2 – a1 = 2

a3 – a2 = 3

.....

an – an-1 = n

and so on.

Now adding all these equations both at the right-hand side, we get;

1 +2+3+4+.....n = 1/2 [n(n+1)]

Whereas on the left-hand side we get;

(a1 – a0) + (a2 – a1)+ (a3 – a2)+ .....+ (an-1 – an-2)+ (an – an-1)

So you can see, all the terms get cancelled but – a0 and an

Therefore, an – a0 = 1/2 [n(n+1)]

or

an = 1/2 [n(n+1)] + a0

Hence, the solution to the recurrence relation with initial condition a0 = 4, is;

an = 1/2 [n(n+1)] + 4

Recursive Algorithm

Recursion is defined as a method of solving problems that involves breaking a problem down into smaller and smaller sub problems until you get to a small enough problem that it can be solved trivially. Simply this is done by the program calling itself to repeat the same steps.

Recursive problems are solved by developing a recursive algorithm. A recursive algorithm is defined as an algorithm which can call itself with smaller (or simpler) input values, and which obtains the result for the current input by applying simple operations to the returned value for

the smaller (or simpler) input. Thus if a problem can be solved by utilizing solutions to smaller versions of the same problem, and the smaller versions reduce to easily solvable cases, then it means that one can use a recursive algorithm to solve that problem.

The recursive method comprises the following:

1. A base case where the recursion stops and begins to unwind, and
2. A recursive step where the method calls itself

A point worth noting is that a recursive algorithm is said to wind up until the base case is reached and then unwinds. During the unwinding process, further statement programs "pending" may be called.

Recursion is a mechanism whereby repetition can be achieved using a method that continues to call itself until some terminating value is reached, i.e. without using some specific repetition construct.

Example

The factorial function "!".

Its definition is:

- 0! = 1
- for all n > 0, n! = n x (n - 1)!

Thus, by repeatedly using the definition, we can work out that

6! = 6 x 5!

= 6 x 5 x 4!

= 6 x 5 x 4 x 3!

= 6 x 5 x 4 x 3 x 2!

= 6 x 5 x 4 x 3 x 2 x 1!

= 6 x 5 x 4 x 3 x 2 x 1 x 1

= 720

Again, notice that the definition of "!" includes both a base case (the definition of 0!) and a recursive part.

Laws of Recursion

Recursive algorithms must obey the following laws:

1. A recursive algorithm must have a base case.
2. A recursive algorithm must change its state and move toward the base case.
3. A recursive algorithm must call itself, recursively.

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An explanation of the laws is as follows:

First law: the base case allows the algorithm to stop recursing, i.e. a problem small enough to solve directly.

Second law: a change of state occurs moving the algorithm towards the base case. This occurs when a change of state takes place and which is best illustrated when a data modification happens.

Finally, third law: the algorithm must call itself.

Example

Find the maximum value in a list of numbers.

The solution to the problem would look like the following:

```
Function find_max(list)
possible_max_1 = first value in list
 possible_max_2 = find_max(rest of the list);
if(possible_max_1 > possible_max_2)
   answer is possible_max_1
else
   answer is possible_max_2
 end
end
```

Example

Adding three numbers.

The solution will be something like

```
function result = add_numbers(a, b, c)
if (nargin() == 2)
   result = a + b;
else if (nargin() == 3)
   result = add_numbers(a+b, c);
else
   error('oops');
 end
end
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Conclusion

This unit introduced the learner to the recursive algorithms. Examples were used to teach the learner how to write recursive algorithms that can solve self repeating functions. The examples used in the teaching of recursion are also not hinged on any computing language and can thus apply across all the languages. Finally the learners were introduced to the laws of recursion.

References:

1. Kenneth H. Rosen. Discrete Mathematics and its applications. Tata McGraw Hill.

2. Kolman, Busby and Ross, Distrete Mathematics Structures, PHI.

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UNIT-3 (DISCRETE MATHEMATICAL STRUCTURE)

LINEAR RECURRENCE RELATIONS WITH CONSTANT COEFFICIENTS

LECTURE NO:-22

Linear Recurrence Relations with Constant Coefficients

Recurrence Relations

A recurrence relation is a functional relation between the independent variable x, dependent variable f(x) and the differences of various order of f (x). A recurrence relation is also called a difference equation, and we will use these two terms interchangeably.

Example1: The equation f (x + 3h) + 3f (x + 2h) + 6f (x + h) + 9f (x) = 0 is a recurrence relation.

It can also be written as

$a_{r+3} + 3a_{r+2} + 6a_{r+1} + 9a_r = 0$

$y_{k+3} + 3y_{k+2} + 6y_{k+1} + 9y_k = 0$

Example2: The Fibonacci sequence is defined by the recurrence relation $a_r = a_{r-2} + a_{r-1}$, $r \geq 2$,with the initial conditions $a_0=1$ and $a_1=1$.

Order of the Recurrence Relation:

The order of the recurrence relation or difference equation is defined to be the difference between the highest and lowest subscripts of f(x) or $a_r=y_k$.

Example1: The equation $13a_r+20a_{r-1}=0$ is a first order recurrence relation.

Example2: The equation 8f (x) + 4f (x + 1) + 8f (x+2) = k (x)

Degree of the Difference Equation:

The degree of a difference equation is defined to be the highest power of f (x) or $a_r=y_k$

Example1: The equation $y^3_{k+3}+2y^2_{k+2}+2y_{k+1}=0$ has the degree 3, as the highest power of y_k is 3.

Example2: The equation $a^4_r+3a^3_{r-1}+6a^2_{r-2}+4a_{r-3} =0$ has the degree 4, as the highest power of a_r is 4.

Example3: The equation $y_{k+3} +2y_{k+2} +4y_{k+1}+2y_k= k(x)$ has the degree 1, because the highest power of y_k is 1 and its order is 3.

Example4: The equation f (x+2h) - 4f(x+h) +2f(x) = 0 has the degree1 and its order is 2.

Linear Recurrence Relations with Constant Coefficients

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A Recurrence Relations is called linear if its degree is one.

The general form of linear recurrence relation with constant coefficient is

$$C_0\, y_{n+r}+C_1\, y_{n+r-1}+C_2\, y_{n+r-2}+\cdots +C_r\, y_n=R(n)$$

Where $C_0,C_1,C_2.....C_n$ are constant and R (n) is same function of independent variable n.

A solution of a recurrence relation in any function which satisfies the given equation.

Linear Homogeneous Recurrence Relations with Constant Coefficients:

The equation is said to be linear homogeneous difference equation if and only if R (n) = 0 and it will be of order n.

The equation is said to be linear non-homogeneous difference equation if R (n) ≠ 0.

Example1: The equation $a_{r+3}+6a_{r+2}+12a_{r+1}+8a_r=0$ is a linear non-homogeneous equation of order 3.

Example2: The equation $a_{r+2}-4a_{r+1}+4a_r= 3r + 2^r$ is a linear non-homogeneous equation of order 2.

A linear homogeneous difference equation with constant coefficients is given by

$$C_0\, y_n+C_1\, y_{n-1}+C_2\, y_{n-2}+\cdots .....+C_r\, y_{n-r}=0 \text{ ..... equation (i)}$$

Where $\mathbf{C_0,C_1,C_2.....C_n}$ are constants.

The solution of the equation (i) is of the form $A\propto_1^K$, where $\propto_1$ is the characteristics root and A is constant.

Substitute the values of $A \propto^K$ for y_n in equation (1), we have

$$C_0\, A\propto^K+C_1\, A\propto^{K-1}+C_2\, A\propto^{K-2}+\cdots ....+C_r\, A\propto^{K-r}=0\text{.....equation (ii)}$$

After simplifying equation (ii), we have

$$C_0\propto^r+C_1\propto^{r-1}+C_2\propto^{r-2}+\cdots C_r=0\text{.....equation (iii)}$$

The equation (iii) is called the characteristics equation of the difference equation.

If $\propto_1$ is one of the roots of the characteristics equation, then $A\propto_1^K$ is a homogeneous solution to the difference equation.

To find the solution of the linear homogeneous difference equations, we have the four cases that are discussed as follows:

Case1: If the characteristic equation has n distinct real roots $\propto_1, \propto_2, \propto_3, \ldots\ldots \propto_n$.

Thus, $\alpha_1^K, \alpha_2^K, \ldots\ldots \alpha_n^K$ are all solutions of equation (i).

Also, we have $A_1 \propto_1^K, A_2 \propto_1^K, \ldots\ldots\ldots A_n \propto_n^K$ are all solutions of equation (i). The sums of solutions are also solutions.

Hence, the homogeneous solutions of the difference equation are

$$y_k = A_1 \propto_1^K, A_2 \propto_2^K, \ldots\ldots\ldots A_n \propto_n^K.$$

Case2: If the characteristics equation has repeated real roots.

If $\propto_1 = \propto_2$, then $(A_1 + A_2 K) \propto_1^K$ is also a solution.

If $\propto_1 = \propto_2 = \propto_3$ then $(A_1 + A_2 K + A_3 K^2) \propto_1^K$ is also a solution.

Similarly, if root $\propto_1$ is repeated n times, then.

$$(A_1 + A_2 K + A_3 K^2 + \ldots\ldots + A_n K_{n-1}) \propto_1^K$$

The solution to the homogeneous equation.

Case3: If the characteristics equation has one imaginary root.

If $\alpha + i\beta$ is the root of the characteristics equation, then $\alpha - i\beta$ is also the root, where α and β are real.

Thus, $(\alpha + i\beta)^K$ and $(\alpha - i\beta)^K$ are solutions of the equations. This implies

$$(\alpha + i\beta)^K A_1 + \alpha - i\beta)^K A_2$$

Is also a solution to the characteristics equation, where A_1 and A_2 are constants which are to be determined.

Case4: If the characteristics equation has repeated imaginary roots.

When the characteristics equation has repeated imaginary roots,

$$(C_1 + C_2 k)(\alpha + i\beta)^K + (C_3 + C_4 K)(\alpha - i\beta)^K$$

Is the solution to the homogeneous equation.

Example1: Solve the difference equation $a_{r-3}a_{r-1}+2a_{r-2}=0$.

Solution: The characteristics equation is given by

s^2-3s+2=0 or (s-1)(s-2)=0

$\Rightarrow$ s = 1, 2

Therefore, the homogeneous solution of the equation is given by

$a_r=C^1{}_r+C^2.2^r$.

Example2: Solve the difference equation $9y_{K+2}-6y_{K+1}+y_K=0$.

Solution: The characteristics equation is

$9s^2-6s+1=0$ or $(3s-1)^2=0$

$\Rightarrow s = \frac{1}{3}$ and $\frac{1}{3}$

Therefore, the homogeneous solution of the equation is given by

$y_K=(C_1+C_2\ k).\left(\frac{1}{3}\right)^K$

Example3: Solve the difference equation $y_K-y_{K-1}-y_{K-2}=0$.

Solution: The characteristics equation is s^2-s-1=0

$$s=\frac{1\pm\sqrt{1+4}}{2}=\frac{1\pm\sqrt{5}}{2}$$

Therefore, the homogeneous solution of the equation is

$$y_K = C_1\left[\frac{1+\sqrt{5}}{2}\right]^K+C_2\left[\frac{1-\sqrt{5}}{2}\right]^K$$

Example4: Solve the difference equation $y_{K+4}+4y_{K+3}+8y_{K+2}+8y_{K+1}+4y_K=0$.

Solution: The characteristics equation is $s^4+4s^3+8s^2+8s+4=0$

$(s^2+2s+2)\ (s^2+2s+2)=0$

s = -1±i,-1±i

Therefore, the homogeneous solution of the equation is given by

$y_K=(C_1+C_2\ K)(-1+i)^K+(C_3+C_4\ K)($

$$a_{r(p)} = \frac{1}{(E^2-4E+4)}.(3r + 2^r) = \frac{1}{(E-2)^2}.(3r) + \frac{1}{(E-2)^2}.2^r$$

$$= 3.\frac{1}{(1-\Delta)^2}(r) + \frac{r(r-1)}{2!}.2^{r-2} = 3(1-\Delta)^{-2}(r) + \frac{r(r-1)}{2!}.2^{r-2}$$

$$=3\,(1+2\Delta).[r]+\frac{r(r-1)}{2!}.2^{r-2}=3(r+2)+r(r-1).\,2^{r-3}$$

$$a_{r(p)} = 3(r+2) + r(r-1).2^{r-3}$$

Therefore, the total solution is $a_r = (C_1 + C_2 r).2^r + 3(r+2) + r(r-1).2^{r-3}$

References:

1. Kenneth H. Rosen. Discrete Mathematics and its applications. Tata McGraw Hill.

2. Kolman, Busby and Ross, Distrete Mathematics Structures, PHI.

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UNIT-3 (DISCRETE MATHEMATICAL STRUCTURE)

PARTICULAR AND TOTAL SOLUTION

LECTURE NO:-23

Particular and Total Solution

Non-Homogeneous Linear Difference Equations and Particular Solution:

There are two methods to find the particular solution of a non-homogeneous linear difference equation. These are as follows:

1. Undetermined coefficients method
2. E and Δ operator method.

1. Undetermined Coefficients Method: This method is used to find a particular solution of non-homogeneous linear difference equations, whose R.H.S term R (n) consist of terms of special forms.

In this method, firstly we assume the general form of the particular solutions according to the type of R (n) containing some unknown constant coefficients, which have to be determined. Then according to the difference equation, we will determine the exact solution.

The general form of a particular solution to be assumed for the special forms of R (n), to find the exact solution is shown in the table.

Form of R (n)	General form to be assumed
Z, here z is constant	A
Z_r, here z is constant	Z_r
P (r), a polynomial of degree n	A_0 rn+A_1 rn-1+··· ..An
Zr. P (r), here P(r) is a polynomial of the nth degree in r. Z is a constant.	[A_0 rn+A_1 rn-1+··· ..An].Z_r

Example1: Find the particular solution of the difference equation $a_{r+2}-3a_{r+1}+2a_r=Z^r$ .....equation (i)

Where Z is some constant.

Solution: The general form of solution is = A. Z^r

Now putting this solution on L.H.S of equation (i), we get

= A $Z^{r+2}-3AZ^{r+1}+2AZ^r=(Z^2-3Z+2)$ A Z^r.....equation (ii)

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Equating equation (ii) with R.H.S of equation (i), we get

$(Z^2-3Z+2)A=1$

$A = \frac{1}{(z-3z+2)} = \frac{1}{(z-1)(z-2)}$ $(Z\neq1, Z\neq2)$

Therefore, the particular solution is $\frac{z^r}{(z-1)(z-2)}$

Example2: Find the particular solution of the difference equation $a_{r+2}-5a_{r+1}+6a_r=5^r$ .....equation (i)

Solution: Let us assume the general form of the solution= A. 5^r.

Now to find the value of A, put this solution on L.H.S of the equation (i), then this becomes

$= A.\ 5^{r+2}-5.A5^{r+1}+6.A5^r$

$= 25A.\ 5^r-25A.5^r+6A.5^r$

$= 6A.5^r$ .....equation (ii)

Equating equation (ii) to R.H.S of equation (i), we get

$A = 1/6$

Therefore, the particular solution of the difference equation is $=(1/6).5^r$.

Example3: Find the particular solution of the difference equation $a_{r+2}+a_{r+1}+a_r=r.2^r$.....equation (i)

Solution: Let us assume the general form of the solution = $(A_0+A_1r).\ 2\text{^}r$

Now, put these solutions in the L.H.S of the equation (i), we get

$= 2^{r+2}\ [A_0+A_1\ (r+2)]+2^{r+1}\ [A_0+A_1\ (r+1)]+2^r\ (A_0+A_1\ r)$

$= 4.\ 2^r\ (A_0+A_1\ r+2A_1\ )+2.2^r\ (A_0+A_1\ r+A_1\ )+2^r\ (A_0+A_1\ r)$

$= r.\ 2^r\ (7A_1\ )+2^r\ (7A_0+10A_1)$.....equation (ii)

Equating equation (ii) with R.H.S of equation (i), we get

$7A_1=1 \quad \therefore A_1=1/7$

$7A_0+10A_1=0 \quad \therefore A_0=-10/49$

Therefore, the particular solution is $2^r\left(\frac{-10}{49}+\frac{1}{7}r\right)$

Solution: The homogeneous solution of this equation is obtained by putting R.H.S equal to zero i.e.,

$a_r-4a_{r-1}+4a_{r-2}=0$

Total Solution

The total solution or the general solution of a non-homogeneous linear difference equation with constant coefficients is the sum of the homogeneous solution and a particular solution. If no initial conditions are given, obtain n linear equations in n unknowns and solve them, if possible to get total solutions.

If $y_{(h)}$ denotes the homogeneous solution of the recurrence relation and $y_{(p)}$ indicates the particular solution of the recurrence relation then, the total solution or the general solution y of the recurrence relation is given by

$y = y_{(h)}+y_{(p)}$.

Example: Solve the difference equation
$a_r-4a_{r-1}+4a_{r-2}=3r+2^r$.....equation (i)

Solution: The homogeneous solution of this equation is obtained by putting R.H.S equal to zero i.e.,
$a_r-4a_{r-1}+4a_{r-2}=0$

The homogeneous solution is $a_{r(h)}= (C_1+C_2\ r).2^r$

The equation (i) can be written as $(E^2-4E+4)\ a_r=3r+2^r$

The particular solution is given as

$$a_{r(p)} = \frac{1}{(E^2-4E+4)}.(3r+2^r) = \frac{1}{(E-2)^2}.(3r) + \frac{1}{(E-2)^2}.2^r$$

$$= 3.\frac{1}{(1-\Delta)^2}(r) + \frac{r(r-1)}{2!}.2^{r-2} = 3(1-\Delta)^{-2}(r) + \frac{r(r-1)}{2!}.2^{r-2}$$

$$=3\ (1+2\Delta).[r]+\frac{r(r-1)}{2!}.2^{r-2}=3(r+2)+r(r-1).\ 2^{r-3}$$

$$a_{r(p)} = 3(r+2) + r(r-1).2^{r-3}$$

Therefore, the total solution is $a_r = (C_1 + C_2r).2^r + 3(r+2) + r(r-1).2^{r-3}$

References:

1. Kenneth H. Rosen. Discrete Mathematics and its applications. Tata McGraw Hill.

2. Kolman, Busby and Ross, Distrete Mathematics Structures, PHI.

UNIT-3 (DISCRETE MATHEMATICAL STRUCTURE)

GENERATING FUNCTION

LECTURE NO:-24

Generating Function

Generating Functions

Generating function is a method to solve the recurrence relations.

Let us consider, the sequence a_0, a_1, a_2....a_r of real numbers. For some interval of real numbers containing zero values at t is given, the function G(t) is defined by the series

$G(t)= a_0, a_1t+a_2 t^2+\cdots +a_r t_r+$.....equation (i)

This function G(t) is called the generating function of the sequence a_r.

Now, for the constant sequence 1, 1, 1, 1.....the generating function is

$$G(t) = \frac{1}{(1-t)}$$

It can be expressed as

$G(t) =(1-t)^{-1}=1+t+t^2 +t^3+t^4+\cdots$ [By binomial expansion]

Comparing, this with equation (i), we get

$a_0=1,a_1=1,a_2=1$ and so on.

For, the constant sequence 1,2,3,4,5,..the generating function is

$G(t) = \frac{1}{(1-t)^2}$ because it can be expressed as

$G(t) =(1-t)^{-2}=1+2t+3t^2 +4t^3+\cdots +(r+1) t^r$

Comparing, this with equation (i), we get
$a_0=1,a_1=2,a_2=3,a_3=4$ and so on.

The generating function of Z^r,($Z\neq0$ and Z is a constant)is given by

$G(t)= 1+Zt+Z^2 t^2+Z^3 t^3+\cdots +Z^r t^r$

$G(t)=\frac{1}{(1-Zt)}$ [Assume $|Zt|<1$]

So, $G(t)=\frac{1}{(1-Zt)}$ generates Z^r,$Z\neq0$

Also,If $a^{(1)}_r$ has the generating function $G_1(t)$ and $a^{(2)}_r$ has the generating function $G_2(t)$, then $\lambda_1 a^{(1)}_r+\lambda_2 a^{(2)}_r$ has the generating function $\lambda_1 G_1(t)+ \lambda_2 G_2(t)$. Here λ_1 and λ_2 are constants

Application Areas:

Generating functions can be used for the following purposes -

- For solving recurrence relations
- For proving some of the combinatorial identities
- For finding asymptotic formulae for terms of sequences

Example: Solve the recurrence relation $a_{r+2}-3a_{r+1}+2a_r=0$

By the method of generating functions with the initial conditions $a_0=2$ and $a_1=3$.

Solution: Let us assume that

$$G(t)=\sum_0^\infty a_r t^r$$

Multiply equation (i) by t^r and summing from r = 0 to ∞, we have

$$\sum_0^\infty a_{r+2} t^r - 3\sum_0^\infty a_{r+1} t^r + 2\sum_0^\infty a_r t^r=0$$

$(a_2+a_3 t+a_4 t^2+\cdots)-3(a_1+a_2 t+a_3 t^2+\cdots)+2(a_0+a_1 t+a_2 t^2+\cdots)=0$

$[\because G(t)=a_0+a_1 t+a_2 t^2+\cdots]$

$$\therefore \quad \frac{G(t)-a_0-a_1 t}{t^2} - 3\left(\frac{G(t)-a_0}{t}\right) +2G(t)=0 \text{.....equation (ii)}$$

Now, put $a_0=2$ and $a_1=3$ in equation (ii) and solving, we get

$$G(t) = \frac{2-3t}{1-3t+2t^2} \text{ or } G(t) = \frac{2-3t}{(1-t)(1-2t)}$$

$$\text{Now, Let } \frac{2-3t}{(1-t)(1-2t)} = \frac{A}{1-t} + \frac{B}{1-2t}$$

$$\text{i.e., } \quad 2-3t = A(1-2t)+B(1-t) \text{.....equation (iii)}$$

Put t=1 on both sides of equation (iii) to find A. Hence

$-1=-A \quad \therefore A = 1$

Put t=$\frac{1}{2}$ on both sides of equation (iii) to find B. Hence

$\frac{1}{2}=\frac{1}{2}$ B $\quad \therefore$ B = 1

Thus G (t) = $\frac{1}{1-t}+\frac{1}{1-2t}$.Hence,$a_r$=1+$2^r$.

References:

1. Kenneth H. Rosen. Discrete Mathematics and its applications. Tata McGraw Hill.

2. Kolman, Busby and Ross, Distrete Mathematics Structures, PHI.

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